

IPMH: Integration of stepped care for perinatal mood and anxiety disorders among women attending MCH clinics

Data and Safety Monitoring Board (DSMB): Interim Analysis

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Person-time calculations

50% expected person-time calculation

Total expected person-time was estimated as the mean time from enrollment to delivery (calculated as mean gestational age at delivery minus mean gestational age at enrollment), plus 14 weeks postpartum, multiplied by the total expected sample size of 2,970 participants, yielding a total expected person-time of 67741.63 weeks. 50% of the expected person-time is 33870.81 weeks.

Current total person-time elapsed to date calculation

As of the data freeze date of 11 May 2026, the total person-time elapsed among enrolled participants is 57715.14 weeks. This represents 85.2% of the expected person-time. Because the current person-time elapsed is more than 50% of the expected person-time, we proceed with the interim analysis as planned.

Interim analysis results

Generalized linear mixed models (GLMM) with a Gaussian family and identity link were used to compare change in PHQ-9 and GAD-7 scores from enrollment to 14 weeks postpartum between arms. Models included fixed effects for treatment arm, time since enrollment, and their interaction, and a random effect for facility.

PHQ-9

```
phq_model <- lmer(phq9_score ~ arm * time + (1 | clt_study_site),  
                  data = analysis_df)
```

```
summary(phq_model)
```

```
## Linear mixed model fit by REML ['lmerMod']  
## Formula: phq9_score ~ arm * time + (1 | clt_study_site)  
## Data: analysis_df  
##  
## REML criterion at convergence: 17893.5  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max   
## -3.0418 -0.6891 -0.0853  0.5715  6.4218   
##  
## Random effects:  
## Groups      Name      Variance Std.Dev.  
## clt_study_site (Intercept) 1.980    1.407  
## Residual              7.507    2.740  
## Number of obs: 3668, groups: clt_study_site, 20  
##  
## Fixed effects:  
##              Estimate Std. Error t value  
## (Intercept)    6.593164   0.451828  14.592  
## armIntervention  0.810461   0.639090   1.268  
## time           -0.144188   0.005981 -24.106  
## armIntervention:time -0.029042   0.008481  -3.424  
##  
## Correlation of Fixed Effects:  
##              (Intr) armInt time  
## armIntrvntn -0.707  
## time        -0.100  0.071  
## armIntrvnt:  0.070 -0.099 -0.705
```

GAD-7

```
gad_model <- lmer(gad7_score ~ arm * time + (1 | clt_study_site),  
                 data = analysis_df)
```

```
summary(gad_model)
```

```
## Linear mixed model fit by REML ['lmerMod']  
## Formula: gad7_score ~ arm * time + (1 | clt_study_site)  
## Data: analysis_df  
##  
## REML criterion at convergence: 17947  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.6562 -0.6995 -0.1113  0.6037  5.5002   
##  
## Random effects:  
## Groups          Name          Variance Std.Dev.  
## clt_study_site (Intercept) 1.910     1.382  
## Residual              7.579     2.753  
## Number of obs: 3672, groups: clt_study_site, 20  
##  
## Fixed effects:  
##              Estimate Std. Error t value  
## (Intercept)    5.732596   0.444048  12.910  
## armIntervention  0.946175   0.628086   1.506  
## time           -0.124244   0.006011 -20.668  
## armIntervention:time -0.030284   0.008516  -3.556  
##  
## Correlation of Fixed Effects:  
##              (Intr) armInt time  
## armIntrvntn -0.707  
## time         -0.102  0.072  
## armIntrvnt:  0.072 -0.102 -0.706
```

O'Brien-Fleming boundary

The O'Brien-Fleming boundary for the current person-time (85.1989343% information time) is 2.195. The boundary is crossed if the absolute value of the Z-statistic exceeds this value.

Table 1: Primary Outcomes: GLMM Results

Outcome	Estimate	95% CI	OBF-Adjusted CI	Z-statistic	Boundary Crossed
PHQ-9	-0.029	(-0.046, -0.012)	(-0.048, -0.010)	-3.424	YES
GAD-7	-0.030	(-0.047, -0.014)	(-0.049, -0.012)	-3.556	YES

